

I need to pitch the paper to programming language theorists who use proof assistants, not category theorists

Start w/ a section describing syntax of programming languages as free categories

Then talk about what categorical structures this freeness properties refer to

Then gluing as an application of freeness

How do we get concrete categories that fit the framework?

↳ Well, just show that they exhibit the universal properties.

↳ Can make it easier by using the local eliminator. Mention reindexing, local-to-global, etc

↳ Problem, performance issue

↳ Show how Eq solves it

Need to fit in the novel presentation of universal properties via universal elements

Wrap up w/ examples showing conservativity and coherence

↳ If we get other examples, like presented categories, they go here

↳ or normalization

Future work

Metatheory for different languages

↳ Gluing for linear λ -calculus

↳ Hyland-Schellk, monoidal Grothendieck in more generality?

↳ Dependent type theory

↳ Call-by-push-value

↳ Effectful languages

↳ Guarded stuff

↳ Could an analytic treatment of cubical type theory be done in Cubical?

Richer theorems

↳ Normalization

operational semantics

Richer semantics

↳ Sheaves

↳ can be used for normalization proofs of languages w/ coproducts

Meeting starts

Need to assume no knowledge of CC/CCCs

"Need to know current state of affairs"

↳ No reuse

↳ ad-hoc

Read "Fuck Nuance"

Need new title

Displayed $\Rightarrow$ Dependent

Freeness $\Leftarrow$ dependent eliminator

↳ just as lists

Largs aren't just ad-hoc inductive constructors

↳ some kind of type-based presentation via reusable unprops

Need a diagram of gluing

Talk how our LRs aren't just prop valued

↳ Meaning canonicity gives a verified interpreter rather than existence of a canonical form

"Intrinsically verified interpreter"

↳ constructed compositionally

Gluing diagrams

↳ not a new idea, but is the right framing of the existing arguments using displayed categories

Section 2: Need to get across that

this is the LR you'd get manually

but the library is doing bookkeeping

Max suggests either

↳ simple ex like free cat in full detail

↳ or richer example in less detail

↳ lean towards this

Going to treat free cat as syn

and going to prove a metatheoretic prop like canonicity

↳ emphasized as verified w/ prop being fully specified